Polish Notation Handout[1]

Polish notation is a translation technique by which an arithmetic expression is converted into equivalent Polish notations called *prefix* and *postfix*. This technique eliminates the need for parenthesis that are usually used with infix expressions to indicate the order of expression execution (based on operator priority). The notation is referred to as *Polish Notation*.

Definition:

Let *O* be a set of operators, and let *P* be a set of operands. The two forms of Polish notation, *Prefix* and *postfix* Polish, are defined recursively as follows:

(1) If an infix expression *E* is a single operand *a* in *P*, then both prefix and postfix Polish representations of *E* are *a*.

(2) If $E_1\ O\ E_2$ is an infix expression, where *O* is an operator, and where E_1 and E_2 are infix expressions, then

   (a) $O\ E_1'\ E_2'$ is the prefix Polish representation of $E_1\ O\ E_2$, where E_1' and E_2' are the prefix representations of E_1 and E_2, respectively, and

   (b) $E_1''\ E_2''\ O$ is the postfix Polish representation of $E_1\ O\ E_2$, where E_1'' and E_2'' are the postfix representations of E_1 and E_2, respectively.

(3) If (*E*) is an infix expression, then

   (a) The prefix Polish representation of (*E*) is the prefix Polish representation of *E*, and

   (b) The postfix Polish representation of (*E*) is the postfix Polish representation of *E*.

Examples: *Fully parenthesize the expression and work it inside out.*

```
1) -(a+b)*c     ====>    0-[(a+b)*c]

     prefix:    0-[(a+b)*c]
                0-[(+ab)*c]
                0-[*+abc]
                -0*+abc
                -*+abc
```

[1] The term "Polish" is used, as this notation was first described by the Polish mathematician Jan Lukasiewicz.

```
     postfix:  0-[(ab+)*c]
               0-[ab+c*]
               0ab+c*-
               ab+c*-
```

2) (a+b)-(c+d*c) ====> (a+b)-(c+[d*c])

```
     prefix:   (a+b)-(c+[*dc])
               (a+b)-(c+*dc)
               (a+b)-(+c*dc)
               (+ab)-(+c*dc)
               -+ab+c*dc


     postfix:  (a+b)-(c+[dc*])
               (a+b)-(c+dc*)
               (a+b)-(cdc*+)
               (ab+)-(cdc*+)
               ab+cdc*+-
```

3) a+a*c-((d+f)/c*(l-k))/g =====> (a+(a*c))-[({(d+f)/c}*(l-k))/g]

```
     prefix:    (+a*ac)-[({+df/c}*(l-k))/g]
                (+a*ac)-[({+df/c}*(-lk))/g]
                (+a*ac)-[({/+dfc}*(-lk))/g]
                (+a*ac)-[(*/+dfc-lk)/g]
                (+a*ac)-[/*/+dfc-lkg]
                -+a*ac/*/+dfc-lkg


     Postfix:   (aac*+)-[({df+/c}*(l-k))/g]
                (aac*+)-[({df+c/}*(l-k))/g]
                (aac*+)-[({df+c/}*(lk-))/g]
                (aac*+)-[(df+c/lk-*)/g]
                (aac*+)-[df+c/lk-*g/]
                aac*+df+c/lk-*g/-
```